

Hat layout in WoolyGraphs: crown shaping and edge-length uniformity

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Abstract

Notes on two questions about the circular-hat pipeline. First, what makes a knitted crown close like a sphere, and how well do our charts do it. Second, how uniform are the edge lengths the layout optimizer produces, measured against the gauge they are supposed to hit. All measurements are at 8 stitches per inch horizontal and 12 rounds per inch vertical gauge. These are single-run measurements of the current code, recorded as a baseline for comparing optimization protocols later, not averaged trials.

1 Setup and notation

A chart is one pattern repeat: rounds in knit order, one cell per stitch slot, blank where no stitch exists. The hat is that chart tiled P times around. Write h for the horizontal gauge in stitches per inch, v for the vertical gauge in rounds per inch, W_0 for the live stitches per repeat at the widest round, and $N_0 = PW_0$ for the total.

The layout is a graph on stitch positions in $\mathbb{R}^3$. Two edge families carry the gauge:

- **horizontal** (ring): each stitch to its right-hand neighbour in the same round, rest length $1/h = 0.125$ in;
- **vertical** (column): each stitch to the stitch above it in its column, rest length $1/v \approx 0.0833$ in.

Column edges with a decrease or a cast-on at either end are counted separately as **shaping** edges. They nominally target $1/v$ as well, but a **k2tog** merges two columns into one, so there is no reason for them to sit at plain column gauge, and mixing them into the vertical bucket smears it.

The initial layout places round r , stitch i of n_r on a helix at fractional turn $r + i/n_r$, at the radius whose chord between neighbouring stitches is exactly $1/h$. The cast-on round is flat at $z = 0$ and is held fixed during optimization, so the fabric hangs from a brim that is gauge-exact by construction.

2 Crown shaping

2.1 The law

A hat closes like a sphere when each round's live stitch count is the circumference of the sphere at that latitude. Taking the sphere radius from the brim,

$$R = \frac{N_0}{2\pi h}, \quad W(r) = W_0 \cos\left(\frac{r - r_{\text{eq}}}{vR}\right), \quad (1)$$

where r_{eq} is the round at which the cylinder ends. Rounds advance by arc length along the surface, $1/v$ per round, so the angle from the equator after $r - r_{\text{eq}}$ rounds is $(r - r_{\text{eq}})/(vR)$. The crown therefore needs

$$\frac{\pi}{2}vR \quad (2)$$

rounds to reach the pole. This is the same relation as the **Sphere Hat Decs** spreadsheet, and it is what the Shaping tab overlays on the chart.

Equation (2) is the constraint that bites in practice. It couples three things a chart author sets independently: the repeat count, the gauge, and how many rounds of chart sit above the cylinder. A crown that is too short for its own circumference cannot be fixed by moving decreases around.

2.2 Method

To ask whether a chart’s crown *is* a sphere, fit equation (1) to the chart’s live-stitch profile with r_{eq} as the only free parameter, and report the root-mean-square residual in stitches per repeat.

The equator must be fitted rather than read off the chart. The obvious estimator, “the last round at full width”, is biased: near the equator the cosine is flat, so a correctly shaped chart holds full width for several rounds past the true r_{eq} , and taking the last of them compresses the remaining crown and manufactures a large residual. On the reshaped **alt_cubes** below, that estimator puts the equator seven rounds too high and reports a residual ten times the fitted one.

2.3 Results

chart	P	cast-on	circ. (in)	crown rounds	needed	rms
small_cubes	4	136	17.0	50	54.0	1.41
alt_cubes , as extracted	6	120	15.0	50	49.5	0.74
alt_cubes , reshaped	6	120	15.0	49	49.5	0.29
alt_cubes , reshaped	7	140	17.5	55	57.7	0.76

Table 1: Best-fit sphere residual, in stitches per repeat. “Crown rounds” is what the chart provides above the fitted equator; “needed” is equation (2).

small_cubes is the reference: at four repeats it tracks the sphere to ± 2.5 stitches per repeat, or about 4% of its repeat width, and it is the chart whose knitted result is known to be good. Its residual is not the smallest in the table, which is worth keeping in mind before treating small residuals as the goal in themselves.

The extracted **alt_cubes** crown fits at rms 0.74, but the residual understates two visible defects. It holds full width about ten rounds too long and then decreases faster than the sphere to catch up, and it reaches one stitch per repeat five rounds before the chart ends, so the last five rounds are a single-stitch spike rather than a closure. Reshaping re-times the same gore geometry onto equation (1), keeping the colours and the column at which the wedge opens: the profile then agrees with the sphere to within ± 0.7 stitches per repeat and closes smoothly (5, 5, 4, 3, 2, 2, 1, 1 over the last eight rounds, against 5, 3, 3, 1, 1, 1, 1, 1 before).

The shaping in a chart is correct for one repeat count only. At $P = 7$ the same reshaped chart is a 17.5 in hat, closer to **small_cubes** in size, but equation (2) then asks for 58 crown rounds where the chart has 55, and the residual roughly triples. That is still better than the chart started, so $P = 7$ is a defensible trade of shape for size; making it exact means re-running the reshape at $P = 7$ and adding about three rounds of crown.

3 Edge-length uniformity

3.1 Method

The layout optimizer minimises a weighted sum of three terms: a gauge term (squared relative error of every edge against its rest length), an inflate term ($1/r$ repulsion from the centroid, which keeps the surface from collapsing), and a smoothness term on the solid angle subtended at each vertex by its neighbour fan.

To measure the result, take the same edge set the optimizer works on and record the realised length of every edge. For `small_cubes` at four repeats that is 16,888 edges: 8,448 horizontal, 8,016 vertical, and 424 shaping. Report per-class mean, median and standard deviation of the lengths, and summarise the whole layout by the root-mean-square of the relative error $\ell_e/\ell_e^* - 1$.

3.2 Results

layout	class	gauge (in)	mean (in)	std (in)	mean vs gauge
initial helix	horizontal	0.1250	0.1252	0.0023	+0.2%
	vertical	0.0833	0.0996	0.0365	+19.5%
	shaping	0.0833	0.1517	0.0504	+82.0%
after 1000 iterations	horizontal	0.1250	0.1279	0.0063	+2.3%
	vertical	0.0833	0.0904	0.0056	+8.5%
	shaping	0.0833	0.0990	0.0084	+19.9%

Table 2: Edge lengths for `small_cubes` at four repeats, before and after 1000 Adam iterations at learning rate 0.002 with weights $\lambda_{\text{gauge}} = 1.0$, $\lambda_{\text{inflate}} = 0.02$, $\lambda_{\text{smooth}} = 0.1$.

Overall relative error falls from 36.8% to 9.1% rms. Three observations.

The helix is exact where it is trivially exact. Horizontal edges start at +0.2% with a standard deviation of 0.0023 in, because the helix places each ring at the radius that makes the chord exactly $1/h$. Vertical edges are exact in the cylinder for the same reason and wrong only in the crown, which is why their initial median is exactly $1/v$ while their mean is +19.5%. A mean alone would have suggested the whole fabric was stretched; the median says the error is confined to the shaping region. Shaping edges start +82% long, since a decrease edge spans two columns in the initial placement.

Optimization trades peak error for spread. Root-mean-square error improves by a factor of four, and the vertical standard deviation tightens by a factor of six, from 0.0365 to 0.0056 in. But the number of edges worse than 10% off gauge *rises*, from 1,761 to 3,010: the optimizer removes a small population of very large errors and distributes a modest error widely. Any comparison between protocols should therefore report a distribution statistic, not a threshold count, which can move the wrong way while the layout improves.

There is a systematic stretch. Every class ends up long: +2.3% on rings, +8.5% on columns, +19.9% on shaping edges. The signs agree across classes and the effect survives further iterations, so this is the inflate term pulling against the gauge term at equilibrium rather than incomplete

convergence. Lowering λ_{inflate} should move all three distributions back onto their targets, and how far it can be lowered before the surface collapses is the obvious next experiment.

3.3 Tooling

The Edges tab plots the three distributions overlaid on shared bins with the gauge for each drawn as a rule, and lists mean, median and standard deviation beneath. The count axis is logarithmic by default: these distributions put thousands of edges in one bin and tens in the tail, and the tail is the part the question is about. The x axis can switch from inches to error relative to gauge, which places all three classes on a single reference line at zero.

The strain heatmap in the main view replaces the stitches with their edges, coloured by signed error against gauge: black on gauge, orange too long, teal too short. On-gauge edges recede into the background, so what remains visible is exactly the problem regions. Both views read the same edge set as the optimizer, so they cannot disagree with it.

4 Caveats and next steps

Every number here is one run of one configuration, on charts of roughly 8,500 stitches. Nothing is averaged over seeds, and the optimizer’s starting point is deterministic, so run-to-run variation is unmeasured.

The rest length assigned to shaping edges is a modelling choice, not a measurement. Calling it $1/v$ makes every decrease edge look 20% long even in a good layout. A `k2tog` spans two columns at the round below, so a rest length nearer $\sqrt{(1/v)^2 + (1/2h)^2} \approx 0.104\text{in}$ may be the honest target; adopting it would change the shaping row of Table 2 substantially and is worth settling before protocols are compared on that class.

For comparing optimization protocols, the measurements to hold fixed are the chart, the repeat count, the gauge, and the iteration budget; the statistics to report are rms relative error overall and per-class standard deviation. The current method is the baseline in Table 2.